EXPERIMENT 8: BACKWARD CHAINING

AIM

To implement backward chaining in Prolog using rules and facts to prove a goal.

PROCEDURE

1. Define the facts required for each problem.

2. Write a rule that derives the target goal from the given facts.

3. Submit the query and observe whether Prolog proves the goal.

CODE

1. Medical Diagnosis

```
% Facts
has_fever(john).
has_cough(john).

% Rule
has_flu(X) :- has_fever(X), has_cough(X).

% Query
?- has_flu(john).
```

OUTPUT

```
true.
```

2. Student Eligibility Checker

```
% Facts
passed(alice, maths).
passed(alice, science).

% Rule
eligible(X) :- passed(X, maths), passed(X, science).

% Query
?- eligible(alice).
```

OUTPUT

```
true.
```

3. Travel Planner

```
% Facts
has_ticket(james).
has_visa(james).

% Rule
can_travel(X) :- has_ticket(X), has_visa(X).

% Query
?- can_travel(james).
```

OUTPUT

```
true.
```

RESULT

Thus, backward chaining was demonstrated successfully using Prolog facts, rules, and queries for medical diagnosis, student eligibility, and travel planning.